



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

1) Scenario

Review the scenario below. Then complete the step-by-step instructions.

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets
- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns

- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF). You will use the CSF to help you navigate through the different steps of analyzing this cybersecurity event and integrate your analysis into a general security strategy. We have broken the analysis into different parts in the template below. You can explore them here:

- Identify security risks through regular audits of internal networks, systems, devices, and access privileges to identify potential gaps in security.
- Protect internal assets through the implementation of policies, procedures, training and tools that help mitigate cybersecurity threats.
- Detect potential security incidents and improve monitoring capabilities to increase the speed and efficiency of detections.
- Respond to contain, neutralize, and analyze security incidents; implement improvements to the security process.
- Recover affected systems to normal operation and restore systems data and/or assets that have been affected by an incident.

Summary	Our organization fell victim to a Denial of Service (DoS) attack where an unconfigured firewall allowed a malicious actor to flood the internal network with ICMP packets. This completely overwhelmed network capacity, rendering internal resources unavailable to legitimate traffic for two hours. The incident management team successfully mitigated the attack by blocking the ICMP flood, isolating non-essential systems, and prioritizing the recovery of critical infrastructure. To prevent a recurrence, the security team is leveraging the NIST Cybersecurity Framework (CSF) to transition from reactive containment to proactive defense. Moving forward, the strategy shifts away from credential and database security to focus strictly on network perimeter hardening, specifically through automated ICMP, source IP spoof verification,
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	and continuous traffic monitoring using integrated IDS/IPS systems.
Identify	The incident response group conducted a comprehensive review of the company's infrastructure, endpoints, and user permissions to pinpoint security vulnerabilities. Their investigation revealed that an attacker compromised an intern's credentials to gain unauthorized entry into the client database, which subsequently led to the deletion of certain customer records.
Protect	To defend against breaches, the team enforced stricter authentication protocols, including multi-factor authentication (MFA), a three-strike account lockout policy, and mandatory credential safety training for the workforce. Furthermore, they plan to deploy an intrusion prevention system (IPS) and update firewall configurations to maximize perimeter defense.
Detect	For early identification of unauthorized access attempts moving forward, the security team will deploy an intrusion detection system (IDS) alongside firewall logging utilities to continuously scan and analyze all inbound web traffic.
Respond	Immediate containment actions included deactivating the compromised profile and conducting credential protection workshops for the staff. Leadership was notified of the breach and is preparing to mail out data breach notifications to affected clients, while also coordinating with law enforcement and regulatory bodies to comply with legal mandates.
Recover	The database will be returned to its original state using the full backup generated the previous evening. Because any customer data modified or logged during the morning hours was not captured in that snapshot,

	employees have been instructed to manually input those specific updates once the system restoration is finalized.
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Reflections/Notes:
